Vertical Transport

A low rise building might be described as one where the able bodied do not need a lift to reach their floor, but if one is available they invariably use it. This would imply a building of 3-5 floors. A midrise building is one where there may be 8-10 floors and the lift becomes essential, in order for occupants to use the building. A high rise building might be one which contains 15-16 floors and maybe equipped with lifts serving two zones.

As a general rule, about 60 floors can be served from a main terminal lobby at ground level, by up to four groups of lifts (a practical limit). If double deck lifts are used, this permits up to 80 floors to be served from a main terminal lobby. Buildings with more than 80 floors require sky lobbies with shuttle lifts to serve them. This permits buildings of 120/160 floors with one sky lobby and buildings of 180/240 floors with two sky lobbies with single/double deck lifts. Remember the maximum practical number of lifts that can be grouped together is eight cars with four facing four.

Rated speed (m/s)	Travel distance to reach rated speed (m)	Time to travel 200 m (s)	Time to travel 400 m (s)
1	1	202	402
2	4	102	202
5	25	45	85
10	100	29	49
20	400	27	37

<table>
<tr><td colspan="2">Step 1
Determine the basic
Building characteristics
as shown here.</td><td colspan="3">Floor-to-floor pitch: 3.5 meters
Floors above main entrance: 17
Tenancy: Multiple
Type of building: Standard office
Population above main entrance: 1600-1700</td></tr>
<tr><td>Step 2
Select the related
handling capacity
factor and
recommended
interval value.</td><td>Type of Building

Prestige office - single tenancy
Prestige office - multiple tenancy
Standard office - single tenancy
Standard office - multiple tenancy</td><td>design factor as % above main entrance population in 5 minutes
17
12
17
12</td><td colspan="2">Recommended interval
In seconds

25
25
30
30</td></tr>
<tr><td colspan="2">Step 3
Calculate the
handling capacity
and limit of lift
travel values.</td><td colspan="3">Handling capacity
= Population above main entrance xHandling capacity design factor/100
= 1600 x 12/100 = 192 (190) people/5 minutes
Limit of lift travel
= Floor-to-floor pitch x Floors above main entrance = 3.5 x 17 = 59.5 meters</td></tr>
<tr><td colspan="2" rowspan="2">Step 4
Select the
recommended
lift speed.</td><td rowspan="2">Speed

0.4
0.63</td><td colspan="2">Recommended limit of travel metres</td></tr>
<tr><td>Hydraulic Drive
3.3
15.0</td><td>Traction Drive
-
15</td></tr>
</table>

	1.0 **1.6** 2.0 2.5 4.0 6.0	20.0 -----	20 **35** 40 50 70+ 100+

Step 5

H = Handling Capacity
(people per 5 minutes)
I = Interval (seconds)
S = Solution No. (in Step 6)

Floors Above Main Entrance	Speed 1.0 m/s			Speed 1.6 m/s		
	H	I	S	H	I	S
4	54 69 77 81 103 116 142 176	33 35 39 22 23 26 27 29	1 2 3 5 6 7 8 9	61 75 83 91 108 112 125 162 197	29 32 36 20 36 21 24 24 26	1 2 3 4 5 6 7 8 9
5	50 75	38 25	1 5	61 69 79 91 103 120 139 170	32 36 39 21 24 25 28 30	1 2 3 5 6 7 8 9
6				55 73 90 107 124 137 150 165 201	35 24 26 28 31 22 34 24 26	1 5 6 7 8 12 9 13 14
7				51 71 87 97 112 126 150 181	37 26 28 31 35 24 26 28	1 5 6 7 8 12 13 14
8				65 80 93 106 115 138 166	27 30 33 23 26 28 30	5 6 7 11 12 13 14

Floors Above Main Entrance	Speed 1.6 m/s			Speed 2.0 m/s			Speed 2.5 m/s			Speed 4.0 m/s		
	H	I	S	H	I	S	H	I	S	H	I	S
9	63 76 84 86 101 109 128 154	29 32 22 35 24 28 30 33	5 6 10 7 11 12 13 14									
11				57 67 76 89 102 102 141	32 36 24 27 29 33 36	5 6 10 11 12 13 14	120 139 156	25 28 33	12 13 14			
14							111 128 142	27 31 36	12 13 14	116 133 145 147 166 174	26 30 21 35 24 18	12 13 15 14 16 18

										184 199 220	28 20 24	17 19 20
17										105 119 129 145 160 173 190	28 32 22 26 31 21 25	12 13 15 16 17 19 20
19										133 149 164 178 195	23 27 32 22 27	15 16 17 19 20

Step 6

From the solution number determine the minimum configuration possible.

Solution No.	Cars in group	Load kg	No. of People
1	2	630	8
2	2	800	10
3	2	1000	13
4	2	1250	16
5	3	630	8
6	3	800	10
7	3	1000	13
8	3	1250	16
9	3	1600	21
10	4	630	8
11	4	800	10
12	4	1000	13
13	4	1250	16
14	4	1600	21
15	5	1000	13
16	5	1250	16
17	5	1600	21
18	6	1000	18
19	6	1250	16
20	6	1600	21

ESCALATOR

(i) Escalator type: Remote or compact

	Compact	*Remote*
(ii) Speed	0.6 m/s to 0.75 m/s	0.6 m/s to 0.75 m/s
(iii) Load cycle	2 h-3.5 h 100% 4 h-2 h 75% 16 h 50%	8 h 100% 12 h 60%
(iv) Design load	1200 N/per step	500 N/per step
(v) Max. truss Deflection	L/1000	L/1500
(vi) Step type	cast aluminium	steel fabricated
vii) Min. Chain Wheel (mm) Min. Step wheel(mm)	100dia×25wide 75dia×25wide	110dia×25wide 75dia×20wide
(viii) Balustrade Material	stainless steel	Aluminium
(ix) Both are remote monitoring		

1.3 Shuttle Lifts (with sky lobbies)

Many tall buildings are divided into several zones: low zone, mid zone, high zone, etc. with service direct from the main terminal floor, situated at ground level. These are called `local' zones. This becomes impractical with very tall buildings and shuttle lifts are employed to take passengers from the ground level main lobby to a `sky lobby'. This could be 200 m (Petronas Towers, Kuala Lumpur, Malaysia).

Passengers disembark at the sky lobby and service is then provided to further low, mid, high zones, etc. using the sky lobby as an upper main terminal floor. The advantage is that the core efficiency is improved, as the hoist ways extend the whole height of the building (except for the intervening equipment spaces) and occupy the same hoist way `footprint'. Shuttle lifts are usually quite large and fast and provide an excellent service to the sky lobby. Their main disadvantage is that the passengers must change lifts mid journey, hence increasing their total journey time.

1.4 Double Decker Lifts

Double deck lifts comprise two passenger cars one above the other connected to one suspension/drive system. The upper and lower decks can thus serve two adjacent floors simultaneously. During peak periods the decks are arranged to serve `even' and `odd' floors respectively with passengers guided into the appropriate deck for their destination. Special arrangements are made at the lobby for passengers to walk up/down a half flight of stairs/escalators to reach the lower or upper main lobby.

One advantage for double deck lifts is that the `hoist way' handling capacity is improved, as effectively there are two lifts in each shaft. A disadvantage for passengers during off peak periods is when one deck may stop for a call with no coincident landing, or car call, required in the other deck. Special traffic control systems, available during off peak periods, attempt to overcome this problem

1.5 Transfer Floors

Most tall and very tall buildings provide some means to travel between zones and stacks. This is sometimes achieved by overlapping zones (PETRONAS Towers), introducing extra stops (Sears Tower) or shuttle lifts (World Trade Center). A common served floor (other than the Main Terminal or Sky Lobby) is important, where there are common facilities to be accessed, e.g.: restaurant, travel bureau, banking, sports facilities, post room, reprographics, etc.

The effect on traffic handling can be disruptive. In general it is important to restrict access to such floors during up peak and down peak, although the purpose of such a floor would be defeated at other times, i.e.: at the mid-day break or during interfloor traffic.

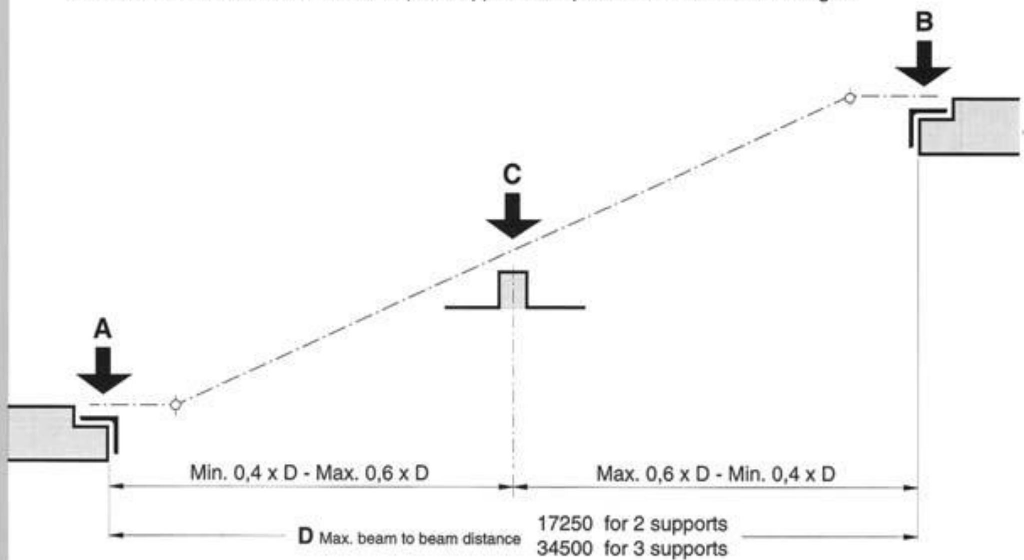
1.6 Top/Down Service

A top/down lift installation is where a sky lobby is used to serve building zones or stacks both in the conventional up direction, but also in the down direction. This does mean that passengers may (psychologically) be concerned that they have travelled up a building only to be then required to travel down to their destination. This technique has only been applied in a few buildings.

Although top/down is more expensive in equipment terms and may complicate the machine room and over travel arrangements, the technique does allow a small footprint building to provide a larger rentable space.

Otis 513 NPE-XL

Manufactured according to Code EN 115.
All details according to the Technical Layout GAA 28300 F.
For special conditions such as extended free span, exterior cladding of more than 200 N/m², wind or earthquake applications please contact IPC-Stadthagen.



Escalator

Single machine	Reaction to support in kN (D in m) ②					
Step width **N**	800			1000		
No. of supports	A	B	C	A	B	C
2	5,07 x D + 6,29	5,07 x D + 12,49	-	5,72 x D + 6,62	5,72 x D + 12,94	-
3	2,02 x D + 4,74	2,02 x D + 11,24	6,33 x D + 2,81	2,30 x D + 5,03	2,30 x D + 11,65	7,16 x D + 2,88

27,3° / 30°

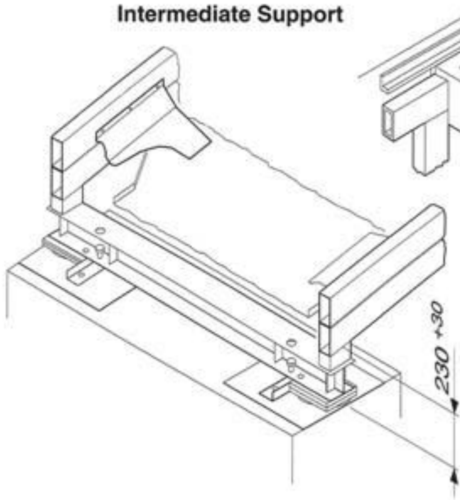
Dual drive	Reaction to support in kN (D in m) ②					
Step width **N**	800			1000		
No. of supports	A	B	C	A	B	C
2	-	-	-	5,72 x D + 7,41	5,72 x D + 20,05	-
3	-	-	-	2,30 x D + 5,03	2,30 x D + 18,36	7,16 x D + 4,07

② For passenger load of 5000 N/m² according to Code EN 115

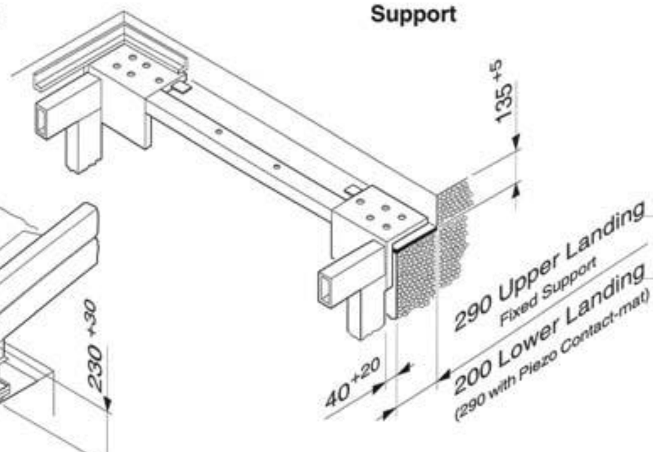
Machine inside truss

Step width:
800/1000 mm

Intermediate Support



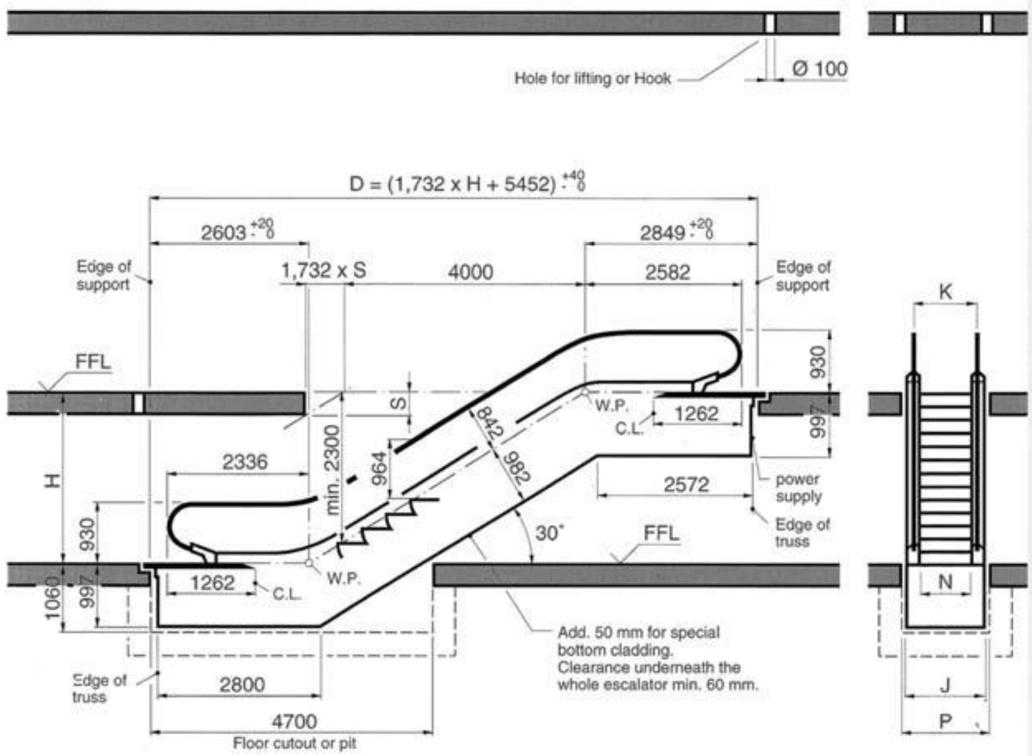
Support



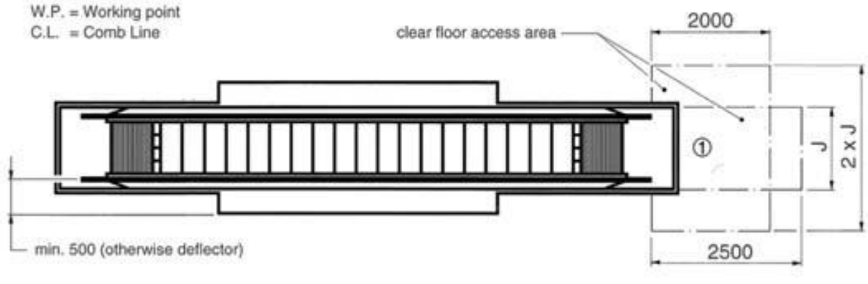
OTIS

RISE 6,001 - 6,5 m

3 flat steps



W.P. = Working point
C.L. = Comb Line



Dimensions in mm for 3 flat steps			
Step width N	**K**	**J**	**P**
800	1005	1347	1430
1000	1208	1550	1630

① Minimum clear floor access area in front of the landings measured from newel end: 2,5 m x escalator width or 2,0 m x twice escalator width.

Otis 506 NCE

Escalator

30°

0,5 m/s

Step width:
800/1000 mm

Rise:
6,001 - 6,5 m

OTIS

1. **Extension beyond comb plates**
Each moving handrail shall extend at normal handrail height not less than 12 in. (305 mm) beyond the line of points of the comb plate teeth at the upper and lower landings.
2. **Balustrades**
Balustrades shall be installed on each side of the escalator.
(a) The balustrade on the step side shall have no areas or moldings depressed or raised more than 1/4 in. (6.4 mm) from the parent surface. Such areas or moldings shall have all boundary edges beveled or rounded.
(b) The balustrade shall be totally closed, except:
Where the handrail enters the newel base;
Gaps between interior panels shall not be wider than 3/16 in. (4.8 mm). The edges shall be rounded or bevelled.
(c) The width between the balustrade interior panels in the direction of travel shall not be changed.
3. **Strength**
Balustrades shall be designed to resist the simultaneous application of a static lateral force of 40 lbf/ft (584 N/m) and a vertical load of 50 lbf/ft (730 N/m), both applied to the top of the handrail stand.
4. **Distance between the handrail centerlines**
The distance b1 between the centerline of the handrails shall not exceed the distance between the skirting by more than 0.45 m.
5. **Protection at the point of entry into the balustrade**
The lowest point of entry of the handrail into the newel shall be at a distance from the floor which shall be not less than 0.10m and not exceed 0.25 m.
The horizontal distance between the furthest point reached by the handrail and the point of entry into the newel shall be at least 0.30 m.
At the point of entry of the handrail into the newel a guard shall be installed to prevent the pinching of fingers and hands.
6. **Height above the steps, pallets and the belt**
The vertical distance between the handrail and step nose pr pallet surface or belt surface shall be not less than 0.90m and not exceed 1.10 m.
7. **Skirt Panels**
(1) The height of the skirt above the tread nose line shall be at least 1in. (25 mm) measured vertically.
(2) Skirt panel shall not deflect more than 1/16 in. (1.6 mm) under a force of 150 lbf (667 N).
8. **Angle of inclination**
Maximum angle to the horizontal in which the steps, the pallets or the belt move. It shall not exceed 30◦ from the horizontal.
9. **Driving machine**
General
Each escalator and each passenger conveyor shall be driven by at least one machine of its own.
(a) Speed
The rated speed of the escalator shall not exceed:
– 0.75 m/s for an escalator with an angle of inclination α up to 30◦;
– 0.50 m/s for an escalator with an angle of inclination α of more than 30◦ up to 35◦.
The rated speed of passenger conveyors shall not exceed 0.75 m/s.
The passenger conveyors are permitted to have a maximum rated speed of 0.90 m/s provided that the width of the pallets or the belt does not exceed 1.10 m, and that, at the landings, the pallets or the belt move horizontally for a length of at least 1.60m before entering the combs.
10. **Stopping distances of the escalator.**

The stopping distances for unloaded and downward moving loaded escalators shall be between the following values.

Rated speed m/s	Stopping distance between (m) Minimum	Maximum
0.50	0.20	1.00
0.65	0.30	1.30
0.75	0.35	1.50

11.**Lighting**

On indoor escalators or passenger conveyors the intensity of illumination shall be not less than 50 lx at the landings; on outdoor escalators or passenger conveyors it shall be not less than 15 lx at the landings, measured at floor level.